Annotated Code Document

By Group 6:

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Project 2 Annotated Code Document

Code	Annotation

Implement sequential User Inputs:

project_no = int(input("Enter project number: ")) control_no = input("Enter control number: ") concrete_class = input("Enter class of concrete: ") date = input("Date: ") ready_mix_plant = input("Contractor: ") target_aggregate_gradation = input("Target Gradation (%): ")	The code allows the user to input project name, control number, concrete class, date, contractor, and target aggregate gradation. These are variables. The project number is the integer type and the rest are string.
weight_cement_A = float(input("Weight of cement per yd3: ")) weight_fly_ash_B = float(input("Weight of fly ash per yd3: ")) weight_silica_fume_C = float(input("Weight of silica fume per yd3: ")) weight_other_scm_D = float(input("Weight of other SCM per yd3: "))	This code allows the user to input the weights of the different components of the mix design. These inputs are all float variables.
target_water_cement_ratio = float(input("Target Water/Cement Ratio: ")) target_percent_air = float(input("Target % Air Content: "))	This code allows the user to input the target water to cement ratio and target percent air. These inputs are both float variables.
target_percent_fine_aggregate = float(input("Target % Fine Aggregate: ")) target_percent_coarse_aggregate = float(input("Target % Coarse Aggregate: ")) target_percent_other_aggregate = float(input("Target % Other Aggregate: "))	This code allows the user to input the target percentage of fine, coarse and other aggregate. These inputs are all float variables.
sg_cement_J = float(input("Enter specific gravity of cement J: ")) sg_fly_ash_K = float(input("Enter specific gravity of fly ash K: ")) sg_silica_fume_L = float(input("Enter specific gravity of silica fume L: ")) sg_other_scm_M = float(input("Enter specific gravity of other SCM M: "))	This code allows the user to input the specific gravities. These inputs are all float variables.

sg_fine_N = float(input("Enter specific gravity of fine aggregate N: ")) sg_coarse_O = float(input("Enter specific gravity of coarse aggregate O: ")) sg_other_P = float(input("Enter specific gravity of other aggregate P: "))	
print("\nAll user inputs collected successfully!")	This prints a statement so that the user knows that all the inputs have been collected.

Translate the Excel Logic into python:

import numpy as np import pandas as pd	This code imports the pandas and numpy package.

cubic_yd_ft3 = 27 #cubic feet in 1 CY
unit_weight_water = 62.4 #lb/ft^3
	This code defined constants that will be used later in the calculation of mix designs

def water_weight(cement_a, fly_ash_b, silica_fume_c, other_scm_d, wc_ratio_e): total_cementitous_weight = cement_a + fly_ash_b + silica_fume_c + other_scm_d water_weight_Q = total_cementitous_weight * wc_ratio_e return water_weight_Q	This code defined a function to calculate water weight using weight of cement, weight of fly ash, weight of silica fume, and other SCM, all per yard.
print(f"water weight, Q = {water_weight(A, B, C, D, E)} lb/yd^3") Q = water_weight(A, B, C, D, E)	In this line the result is calculated and printed using the variables defined. These variables will be taken from the user input section.

def vol_cement(w_cement, sg_cement):
   vol_R = w_cement / (sg_cement * unit_weight_water)
   return vol_R
	This code defines a function to calculate the volume of cement (R)

R = vol_cement(A, J) print(f"cement volume, R = {R:.3} ft^3")	Here the cement weight is calculated and printed using the defined variables. These variables will be taken from the user input section, as well. Then the cement weight calculated is printed.

def fly_ash_vol(w_flyash, sg_flyash):
   vol = w_flyash /(sg_flyash * unit_weight_water)
   return vol
	This defines a function to calculate the fly ash volume.

S = fly_ash_vol(B, K) print(f"fly ash volume, S = {S:.3} ft^3")	The fly ash volume is calculated from the user input variables and printed.

def silica_vol(w_silica, sg_silica):
   vol = w_silica /(sg_silica * unit_weight_water)
   return vol
	This defines a function to calculate the silica fume volume.

T = fly_ash_vol(C, L) print(f"silica fume volume, T = {T:.3} ft^3")	The silica fume volume is calculated from the user input variables and printed.

def scm_vol(w_scm, sg_scm):
   vol = w_scm /(sg_scm * unit_weight_water)
   return vol
	This defines a function to calculate the other SCM volume.

U = scm_vol(D, M) print(f"Other SCM volume, U = {U:.3} ft^3")	The other SCM volume is calculated from the user input variables and printed.
def air_vol(target_air_content): return (target_air_content / 100) * 27	This defines a function to calculate the air volume.
V = air_vol(F) print(f"air volume, V = {V:.3} ft^3")	The air volume is calculated from the user input variables and printed.
def water_vol(weight_water): return weight_water / 62.4	This defines a function to calculate the water volume.
W = water_vol(Q) print(f"water volume, W = {W:.3} ft^3")	The water volume is calculated from the user input variables and printed.
def agg_vol(R, S, T, U, V, W): return 27 - R - S - T - U - V - W	Here a function is defined to calculate the aggregate volume using the volumes previously calculated.
X = agg_vol(R, S, T, U, V, W) print(f"aggregate volume, X = {X:.3} ft^3")	The aggregate volume is calculated from the user input variables and printed.

def fine_aggregate_Y(percent_fine_G, sg_fine_N, volume_X):
   weight_Y = unit_weight_water * (percent_fine_G /100) * sg_fine_N * volume_X
   return weight_Y
	The code defines a function to calculate the fine aggregate weight.

Y = fine_aggregate_Y(percent_fine_G, sg_fine_N, volume_X) print(f"fine aggregate weight, Y = {Y:.3f} lb")	The fine aggregate weight is calculated from the user input variables and printed.
def course_aggregate_Z(percent_coarse_H, sg_coarse_O, volume_X): return unit_weight_water * (percent_coarse_H /100) * sg_coarse_O * volume_X	The code defines a function to calculate the coarse aggregate weight.
Z = coarse_aggregate_Z(percent_coarse_H, sg_coarse_O, volume_X)	The coarse aggregate weight is calculated from the user input variables and printed.

print(f"coarse aggregate weight, Z = {Z:.1f} lb")	
def other_aggregate_AA(percent_other_I, sg_other_P, volume_X): return unit_weight_water * (percent_other_I /100) * sg_other_P * volume_X	The code defines a function to calculate the other aggregate weight.
AA = other_aggregate_AA(percent_other_I, sg_other_P, volume_X) print(f"Other aggregate weight, AA = {AA:.1f} lb")	The other aggregate weight is calculated from the user input variables and printed.

Generate a Final Mix Design Output:

print("\n--")	This prints a line of code. This allows the results to be more organized, similar to an Excel table